

Online Appendix: Rain or Shine? Optimal Utility Pricing under Different Weather Patterns

I. Appendix

All plots of this paper have utilized the software “Colorgorical” to choose a more clear and aesthetically pleasing color scheme Gramazio, Laidlaw and Schloss (2017).

All codes producing the results and graphs related to this project will be made available upon publication.

A. Water Utility Pricing in Austin, TX

PRECIPITATION TREND IN AUSTIN

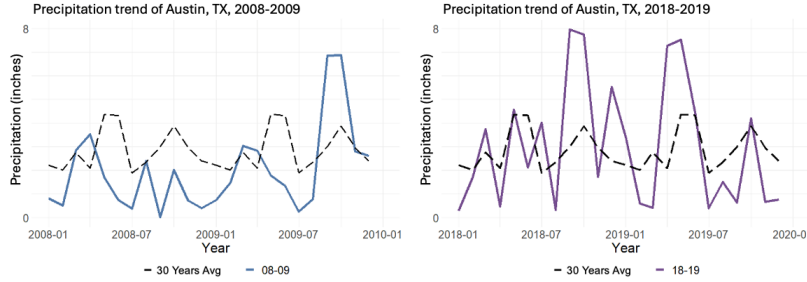


Figure 1. : Precipitation Trends of 2008-2009 vs. 2018-2019

TIER AND INCOME DISTRIBUTION

Based on the status quo price and status quo quantity distribution, I divide the household’s monthly income into 5 different strata. Since both the quantity and income are very right-skewed, only for this plot, I filtered out all households with monthly household income bigger than \$250,000, and quantities higher than 100,000 gallons. These households are not filtered out in demand estimation and counterfactual analysis, but are simply filtered out for the picture below for visualization purposes.

DESCRIPTIVE AND REDUCED FORM EVIDENCE

In the interest of seeing how different income strata will react to abnormal weather changes and the descriptive evidence illustrated in Section II of the main text (Figure 2), I performed the following OLS:

$$\Delta q = \beta_0 + \beta_1 \Delta Z_T + \beta_2 \Delta Z_P + \beta_3 \text{Income} * \Delta Z_T + \beta_4 \text{Income} * \Delta Z_P + \alpha p + \varepsilon$$

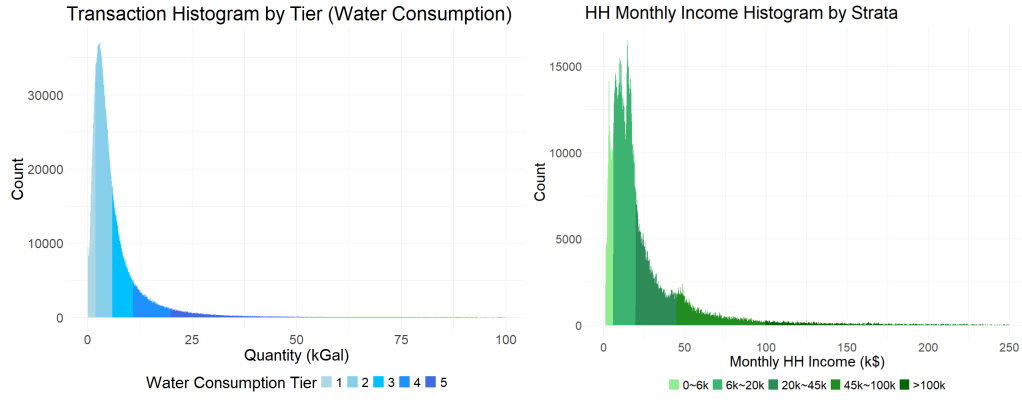


Figure 2. : Monthly Water Quantity Distribution vs. Income Distribution

Table 1—: Monthly Water Quantity Distribution vs. Income Distribution

Income Strata (\$)	Percentage	Tiers (kGal)	Percentage
0 ~ 6k	0.146	0 ~ 2k	0.159
6k ~ 20k	0.489	2k ~ 6k	0.503
20k ~ 45k	0.216	6k ~ 11k	0.200
45k ~ 100k	0.104	11k ~ 20k	0.0927
> 100k	0.0441	> 20k	0.0456

Note: Just like the utility targeting different quantity levels of households. This division of household monthly income into different strata is without loss of generality and serves as a way to gauge the distributional effect incurred by any price change.

where Income is the income strata defined in Table 1. ΔZ for both precipitation and temperature is the difference between observed Z_m for month m and the corresponding 30 year average. $\Delta q = \frac{q_m - \bar{q}_m}{\bar{q}_m}$, which is the quantity deviation from the average quantity for the specific household for a specific month between 2016-2020 in percentage terms.

Table 2—: Regression Results for Quantity Deviation

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	−0.4182	0.0008490	−492.609	< 2e-16 ***
ΔZ_P	−0.01668	0.0003582	−46.555	< 2e-16 ***
ΔZ_T	0.004787	0.0001438	33.283	< 2e-16 ***
income_strata: 6k~20k	−0.02104	0.0007867	−26.739	< 2e-16 ***
income_strata: 20k~45k	−0.08237	0.0008990	−91.627	< 2e-16 ***
income_strata: 45k~100k	−0.1673	0.001096	−152.710	< 2e-16 ***
income_strata: >100k	−0.2476	0.001513	−163.593	< 2e-16 ***
p	0.07486	0.00008868	844.089	< 2e-16 ***
$\Delta Z_P \times \text{income_strata: 6k} \sim 20\text{k}$	0.001576	0.0004098	3.846	0.00012 ***
$\Delta Z_P \times \text{income_strata: 20k} \sim 45\text{k}$	−0.009682	0.0004614	−20.986	< 2e-16 ***
$\Delta Z_P \times \text{income_strata: 45k} \sim 100\text{k}$	−0.01923	0.0005504	−34.933	< 2e-16 ***
$\Delta Z_P \times \text{income_strata: >100k}$	−0.01874	0.0007744	−24.198	< 2e-16 ***
$\Delta Z_T \times \text{income_strata: 6k} \sim 20\text{k}$	0.00009410	0.0001657	0.568	0.57012
$\Delta Z_T \times \text{income_strata: 20k} \sim 45\text{k}$	0.0004584	0.0001880	2.439	0.01474 *
$\Delta Z_T \times \text{income_strata: 45k} \sim 100\text{k}$	−0.002830	0.0002234	−12.667	< 2e-16 ***
$\Delta Z_T \times \text{income_strata: >100k}$	−0.002834	0.0003005	−9.428	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4014 on 2351610 degrees of freedom

Multiple R-squared: 0.263, Adjusted R-squared: 0.263

F-statistic: 5.596e+04 on 15 and 2351610 DF, p-value: < 2.2e-16

In this model, controlling for differences in precipitation, maximum temperature, and the marginal price, the relationship between income strata and the percentage difference in water consumption is seen through: 1) for baseline differences (when Δ weather and price are zero), compared to the lowest income strata (0-6k), most higher income strata show a statistically significant lower baseline percentage difference in water consumption from their usual amount. This does not mean higher income groups use less water overall; it means their deviation from their own usual quantity, under these specific baseline conditions, is lower than the deviation for the reference group. This effect is likely influenced by the strong role of price in this model, as marginal price might be correlated with income or consumption levels that influence which price tier is reached. 2) The linear effect of precipitation difference: the interaction terms between income strata and ΔZ_P are largely significant. This means that the linear rate at which Δq changes for every unit increase in precipitation difference varies significantly across income strata. In particular, higher income strata (20k-45k, 45k-100k, >100k) show increasingly more

negative interaction terms (-0.009682, -0.01923, -0.01874). This means the negative impact of the additional precipitation difference on consumption deviation is increasingly stronger as income rises in these groups. 3) the linear effect of temperature difference: many interaction terms between income strata and ΔZ_T are also significant, indicating that the linear rate at which Δq changes for every unit increase in temperature difference varies significantly across income strata. However, the relationship is less clear-cut compared to the one from precipitation. Compared to the reference group (where the effect of ΔZ_T is 0.004787, meaning a 0.48 percentage point increase in consumption difference for every unit increase in temperature difference), the 6k-20k stratum's interaction is not significant (0.0000941), suggesting its linear temperature sensitivity is not statistically different from the reference group in this model. The 20k-45k stratum has a significant positive interaction (0.0004584), making its positive temperature effect slightly stronger ($0.004787 + 0.0004584 = 0.0052454$). The 45k-100k and >100k strata have significant negative interaction terms (-0.002830 and -0.002834). This means that although the base effect of temperature difference is positive, the additional positive effect seen in higher income groups in the first model is now appearing as a reduction in the sensitivity compared to the reference group's base sensitivity ($0.004787 - 0.002830 = 0.001957$ for 45k-100k; $0.004787 - 0.002834 = 0.001953$ for >100k).

This difference between temperature and precipitation prompts the research to be more focused on changing precipitation. In addition, a structural model is used to estimate the demand on the panel data to show how price, precipitation, and the interaction term with income strata will affect the demand.

B. Demand Model and Estimation

ADDITIONAL CASE IN THE UNCONDITIONAL INDIRECT UTILITY FUNCTION

When constructing the unconditional indirect utility function from the conditional ones, the case where consumption occurs exactly at a tier boundary (kink point), rather than strictly within a tier, may require clarification. Continuing with the two-tier scenario ($K = 2$) without loss of generality, this case arises when the household's optimal consumption calculated using the tier 1 price, $g(p_1, I)$, would exceed the tier 1 limit q_1 , *and* the optimal consumption calculated using the tier 2 price and virtual income, $g(p_2, I + d_2)$, would fall below q_1 . That is, the condition is $g(p_2, I + d_2) \leq q_1 < g(p_1, I)$ (assuming $d_1 = 0$ or defined appropriately). To satisfy the Incentive Compatibility (IC) constraint (i.e., ensure the chosen consumption is utility-maximizing given the full budget set), the household optimally consumes exactly at the kink point q_1 . The diagram below illustrates this:

As shown, under the condition $g(p_2, I + d_2) \leq q_1 < g(p_1, I)$, the optimal (incentive compatible) choice for this household is to consume the bundle corresponding to the kink point, $(q_1, I - \text{Bill}(q_1))$. Here, $\text{Bill}(q_1)$ represents the total water bill incurred when consuming exactly q_1 units (e.g., typically $A_2 + p_1 q_1$ in a two-tier system where A_2 is the fixed charge associated with entering tier 2). If consumption at the kink were not allowed as an option in the model, such consumers would be incorrectly assigned to

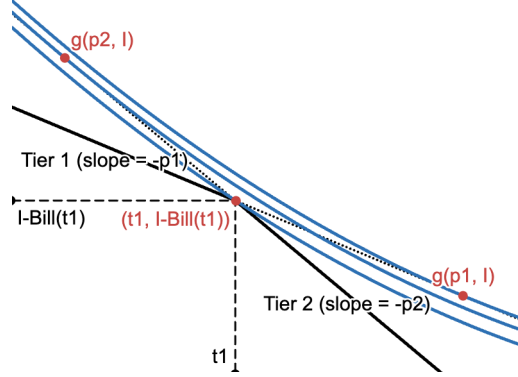


Figure 3. : The household consumes exactly at the kink $(q_1, I - \text{Bill}(q_1))$

consume within one of the adjacent tiers, violating the true optimum.

Generalizing, for any tier boundary q_k where $k \in \{1, \dots, K-1\}$, some households might satisfy the condition $g(p_{k+1}, I + d_{k+1}) \leq q_k < g(p_k, I + d_k)$, for whom the optimal consumption point is exactly $(q_k, I - \text{Bill}(q_k))$. In our empirical application to Austin, which has 5 tiers ($K = 5$), there are 4 such kink points (q_1, q_2, q_3, q_4) , requiring the model to allow for consumption exactly at these quantities to ensure incentive compatibility across the full range of consumption.

LIKELIHOOD FUNCTION

As explained in Section III.A of the main text, the unobservables have two terms $\varepsilon = \eta + v$. $\eta \sim N(0, \sigma_\eta^2)$ represents the household-level heterogeneity observed by households (e.g., preferences), while $v \sim N(0, \sigma_v^2)$ represents the household-level perception/optimization error that is unobserved by households even ex-post. Given the structure of the error term, the unconditional ex post water demand after the realization

of the error term is:

$$(1) \quad \log(w) = \begin{cases} \log(w_1) + \eta + v & \text{if } \eta \leq \log(q_1) - \log(w_1) \\ \log(q_1) + v & \text{if } \log(q_1) - \log(w_1) < \eta \leq \log(q_1) - \log(w_2) \\ \log(w_2) + \eta + v & \text{if } \log(q_1) - \log(w_2) < \eta \leq \log(q_2) - \log(w_2) \\ \log(q_2) + v & \text{if } \log(q_2) - \log(w_2) < \eta \leq \log(q_2) - \log(w_3) \\ \dots & \\ \log(q_{K-1}) + v & \text{if } \log(q_{K-1}) - \log(w_{K-1}) < \eta \leq \log(q_{K-1}) - \log(w_K) \\ \log(w_K) + \eta + v & \text{if } \log(q_{K-1}) - \log(w_K) < \eta \end{cases}$$

This unconditional demand functional form is for a general case of number of total tiers (K). Without loss of generality, I explain the derivation of the likelihood function for a two-tier case ($K = 2$); the approach generalizes to K tiers. Conditional on the ex-ante optimal choice (based on the known preference shock η but ignoring the ex-post error v) being tier 1 ($k^* = 1$), the condition is $\log w_1 + \eta \leq \log q_1$. Similarly, $k^* = 2$ if $\log w_2 + \eta > \log q_1$. The remaining possibility, ensuring incentive compatibility and covering all η , is consumption at the kink q_1 , which occurs if $\log w_2 + \eta \leq \log q_1 < \log w_1 + \eta$. Rearranging these conditions on η and adding the ex-post error v (unobserved by the household when choosing the tier/kink), the observed log-consumption $\log(w)$ is:

$$\log(w) = \begin{cases} \log(w_1) + \eta + v & \text{if } \eta \leq \log(q_1) - \log(w_1) \\ \log(q_1) + v & \text{if } \log(q_1) - \log(w_1) < \eta \leq \log(q_1) - \log(w_2) \\ \log(w_2) + \eta + v & \text{if } \eta > \log(q_1) - \log(w_2) \end{cases}$$

which is the two-tier special case of Equation 7 from the main text. Note that since the household does not observe v when making its choice, v affects the final observed consumption in all scenarios.

Let $f(\cdot)$ denote a probability density function (pdf). The overall likelihood for an observation $\log w$ is the sum of the contributions from these three mutually exclusive and exhaustive scenarios ($L = L_1 + L_{\text{kink}} + L_2$).

Case 1: $k^* = 1$ ($\eta \leq \log(q_1) - \log(w_1)$) The observed log consumption is $\log w = \log w_1 + \eta + v$. The contribution to the likelihood depends on the joint pdf of $(\eta + v, \eta)$, integrated over the relevant range of η :

$$L_1 = \int_{-\infty}^{\log q_1 - \log w_1} f_{v+\eta, \eta}(\log w - \log w_1, \eta) d\eta$$

Case 2: $k^* = 2$ ($\eta > \log(q_1) - \log(w_2)$) The observed log consumption is $\log w = \log w_2 + \eta + v$. The likelihood contribution is:

$$L_2 = \int_{\log q_1 - \log w_2}^{\infty} f_{v+\eta, \eta}(\log w - \log w_2, \eta) d\eta$$

Case 3: Kink Consumption ($\log(q_1) - \log(w_1) < \eta \leq \log(q_1) - \log(w_2)$) The observed log consumption is $\log w = \log q_1 + v$. The likelihood contribution depends on the joint pdf of (v, η) , integrated over the relevant range of η :

$$L_{kink} = \int_{\log q_1 - \log w_1}^{\log q_1 - \log w_2} f_{v,\eta}(\log w - \log q_1, \eta) d\eta$$

If I assume $\eta \sim N(0, \sigma_\eta^2)$ and $v \sim N(0, \sigma_v^2)$, and that they are independent, these integrals can be solved in closed form. Let $\phi(\cdot)$ and $\Phi(\cdot)$ be the standard normal pdf and cdf, respectively. The joint distribution of $(\eta, v + \eta)$ is bivariate normal. Let $\rho_s = \text{Corr}(\eta, v + \eta) = \sigma_\eta / \sqrt{\sigma_\eta^2 + \sigma_v^2}$.

Evaluating the first integral (L_1):

$$\begin{aligned} L_1 &= \int_{-\infty}^{\log q_1 - \log w_1} f_{v+\eta}(\log w - \log w_1) f_{\eta|v+\eta}(\eta | \log w - \log w_1) d\eta \\ &= f_{v+\eta}(\log w - \log w_1) \int_{-\infty}^{\log q_1 - \log w_1} f_{\eta|v+\eta}(\eta | \log w - \log w_1) d\eta \\ &= \frac{1}{\sqrt{\sigma_\eta^2 + \sigma_v^2}} \phi\left(\frac{\log w - \log w_1}{\sqrt{\sigma_\eta^2 + \sigma_v^2}}\right) \Phi\left(\frac{(\log q_1 - \log w_1)/\sigma_\eta - \rho_s(\frac{\log w - \log w_1}{\sqrt{\sigma_\eta^2 + \sigma_v^2}})}{\sqrt{1 - \rho_s^2}}\right) \\ &\equiv \frac{\phi(s_1)}{\sqrt{\sigma_\eta^2 + \sigma_v^2}} \Phi(r_1) \end{aligned}$$

where $s_1 = (\log w - \log w_1) / \sqrt{\sigma_\eta^2 + \sigma_v^2}$ and $r_1 = (t_1^* - \rho_s s_1) / \sqrt{1 - \rho_s^2}$ with $t_1^* = (\log q_1 - \log w_1) / \sigma_\eta$.

Similarly, evaluating the integral for the second case (L_2):

$$\begin{aligned} L_2 &= f_{v+\eta}(\log w - \log w_2) \int_{\log q_1 - \log w_2}^{\infty} f_{\eta|v+\eta}(\eta | \log w - \log w_2) d\eta \\ &= f_{v+\eta}(\log w - \log w_2) \left[1 - \Phi\left(\frac{(\log q_1 - \log w_2)/\sigma_\eta - \rho_s(\frac{\log w - \log w_2}{\sqrt{\sigma_\eta^2 + \sigma_v^2}})}{\sqrt{1 - \rho_s^2}}\right) \right] \\ &\equiv \frac{\phi(s_2)}{\sqrt{\sigma_\eta^2 + \sigma_v^2}} (1 - \Phi(n_2)) \end{aligned}$$

where $s_2 = (\log w - \log w_2) / \sqrt{\sigma_\eta^2 + \sigma_v^2}$ and $n_2 = (m_1 - \rho_s s_2) / \sqrt{1 - \rho_s^2}$ with $m_1 = (\log q_1 - \log w_2) / \sigma_\eta$.

Finally, evaluating the integral for the third case (kink consumption, L_{kink}), using the

independence of v and η :

$$\begin{aligned} L_{kink} &= \int_{t_1^* \sigma_\eta}^{m_1 \sigma_\eta} f_v(\log w - \log q_1) f_\eta(\eta) d\eta = f_v(\log w - \log q_1) \int_{t_1^* \sigma_\eta}^{m_1 \sigma_\eta} f_\eta(\eta) d\eta \\ &= \frac{1}{\sigma_v} \phi\left(\frac{\log w - \log q_1}{\sigma_v}\right) \left[\Phi\left(\frac{m_1 \sigma_\eta}{\sigma_\eta}\right) - \Phi\left(\frac{t_1^* \sigma_\eta}{\sigma_\eta}\right) \right] \\ &= \frac{\phi(u_1)}{\sigma_v} (\Phi(m_1) - \Phi(t_1^*)) \end{aligned}$$

where $u_1 = (\log w - \log q_1) / \sigma_v$.

Summing the three components gives the likelihood for the two-tier case:

$$\begin{aligned} L &= L_1 + L_{kink} + L_2 \\ &= \frac{\phi(s_1)}{\sqrt{\sigma_\eta^2 + \sigma_v^2}} \Phi(r_1) + \frac{\phi(u_1)}{\sigma_v} (\Phi(m_1) - \Phi(t_1^*)) + \frac{\phi(s_2)}{\sqrt{\sigma_\eta^2 + \sigma_v^2}} (1 - \Phi(n_2)) \end{aligned}$$

Using the general definitions $t_k^* = (\log q_k - \log w_k) / \sigma_\eta$, $m_k = (\log q_k - \log w_{k+1}) / \sigma_\eta$, $s_k = (\log w - \log w_k) / \sqrt{\sigma_\eta^2 + \sigma_v^2}$, $u_k = (\log w - \log q_k) / \sigma_v$, $r_k = (t_k^* - \rho_s s_k) / \sqrt{1 - \rho_s^2}$, $n_k = (m_{k-1} - \rho_s s_k) / \sqrt{1 - \rho_s^2}$, and applying boundary conditions ($m_0 \rightarrow -\infty \implies n_1 \rightarrow -\infty \implies \Phi(n_1) = 0$; $q_2 \rightarrow \infty \implies t_2^* \rightarrow \infty \implies r_2 \rightarrow \infty \implies \Phi(r_2) = 1$), this derived likelihood function matches the general form given in Equation 8 of the main text for $K = 2$.

NDVI

The use of the Normalized Difference Vegetation Index (NDVI) for estimating water demand was introduced by Wolak (2016). The essential idea is to quantify vegetation health for specific geographical locations – in this case, the health of lawns associated with single-family homes. Higher NDVI values indicate healthier vegetation, suggesting greater lawn care efforts.

NDVI is calculated from satellite imagery using the difference between the reflectance in the near-infrared (NIR) and red light bands. Healthy plants typically reflect more NIR light and absorb more red light (appearing green), while less healthy or stressed plants reflect less NIR and more red light (appearing yellow or brown). The index is typically calculated as $(\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$, normalized to a range between -1 and +1, where higher values indicate healthier, denser vegetation. Values near -1 often correspond to water bodies. Values around 0 typically represent bare soil or sparse vegetation, and values approaching +1 indicate dense, healthy vegetation like forests or well-maintained lawns.

The raw data I use is the Sentinel-2 Surface Reflectance dataset,¹ providing imagery with a 10m x 10m spatial resolution (pixel size). Since typical residential lots in Texas

¹ If using Google Earth Engine to access data, please refer to Google Earth Engine Sentinel-2 Surface Reflectance.

are larger than this pixel size, this higher resolution allows for a more precise calculation of NDVI within each lot compared to the 30m x 30m resolution used by Wolak (2016). Sentinel-2 images are collected frequently (on average, every 5 days for a given location), though not always regularly, resulting in multiple images per month. Some images, however, contain areas obscured by clouds. To address this, I employ cloud masking and create monthly composite images.² The cloud masking algorithm utilizes bands of the images, such as the “Cloud Probability” or “QA”. These bands indicate the likelihood of cloud cover or contain bitwise flags for clouds and shadows. To create a cloud-free composite, I apply these masks to filter out cloudy pixels from multiple images over all images of a calendar month of the same area, then merge the remaining clear-sky pixels into a seamless, cloud-free composite.

To visually illustrate the process of calculating NDVI in Figure 4, I take a small sample from the Brentwood neighborhood in Austin, near the intersection of W Koenig Lane and Burnet Road. While detecting subtle vegetation differences can be difficult in the raw satellite image, the corresponding NDVI visualization clearly showcases the variations in vegetation health between households.

SUMMARY STATISTICS

After matching the panel data with TCAD records, filtering outliers, and removing households located outside Travis County (and thus likely outside the Austin Water service area), the final dataset primarily covers the period from May 2018 to December 2019. It is important to note that this data includes households on slightly different payment plans. Specifically, some households meeting certain income requirements are eligible for CAP, which offers lower marginal prices. The inclusion of CAP households provides valuable price variation for model identification, despite their small number. The summary statistics can be found in Table 3.³

PRICE ELASTICITIES OF INCOME STRATA

From Section III.C of the main text, I have shown the counter-intuitive result that high-income households are more elastic when the weather is dry. For high elasticity households to pose a “threat” to the pricing design, they will need to be high quantity users as well. I then plot the price elasticity on quantity for all strata.

From Figure 5, there are plenty of higher-income, high elasticity households with large quantity. In the highest income stratum, 14.4% of the households have the elasticity lower than the overall median and have the quantity to be larger than 20k gallons. In addition, there are a decent number of lower-income households that have pretty high consumption levels with not very high elasticities. Both the mismatches between income and quantity, and income and elasticity, explain the reduced-form evidence from Figure 2 of the main text, and weather is one of the factors that causes these mismatches. This

²Details for cloud masking in Google Earth Engine can be found here [Cloud Masking - gee map](#).

³Overall, approximately 60% of households and 55.84% of transactions from the original data are used for the demand estimation analysis. Among the final sample, 99.57% are non-CAP consumers, and 0.43% are CAP consumers.

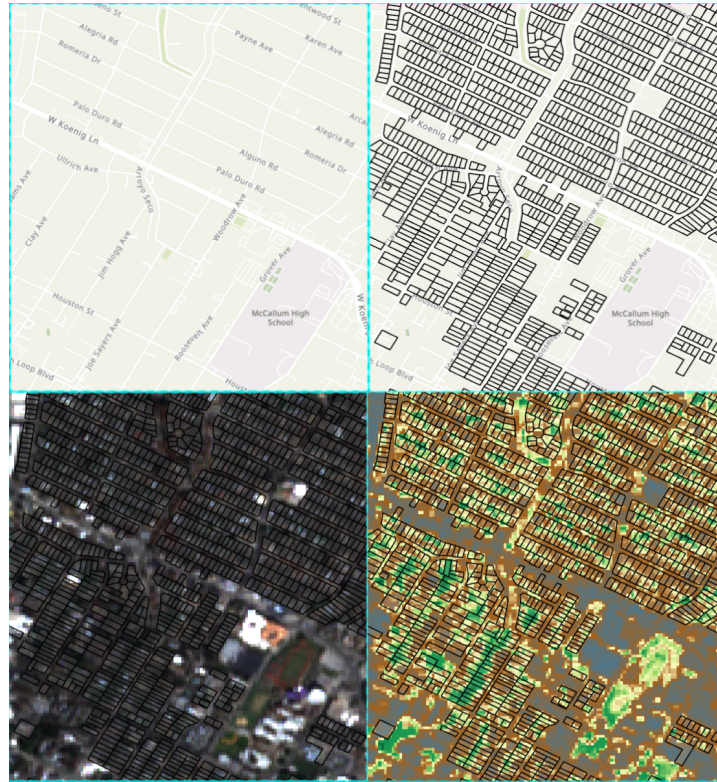


Figure 4. : The Process of Calculating NDVI

Note: From Top Left to Bottom Right: OpenStreetMap base map; OpenStreetMap with household lot shapefiles overlaid; Sentinel-2 Raw Image mosaic with shapefiles overlaid; and final NDVI visualization derived from Sentinel-2 with shapefiles overlaid.

Table 3—: Summary Statistics

Parameter Name	Min	1st Quartile	Median	Mean	3rd Quartile	Max	N
Heavy Water Appliances	0	0	0	0.1637	0	12	127320
Bedrooms	1	1	1	1.439	1	31	127320
Bathrooms	0.1	2	2	2.427	2.5	34	127320
Lot Size (Acre)	0.02136	0.15018	0.19060	0.27396	0.25309	278.25085	127320
Household Monthly Income (k\$)	1.2	8.642	15.320	29.295	27.765	5057.051	127320
NDVI	-0.3564	0.3266	0.4038	0.3988	0.4766	0.7729	2351626
Mean Max Temp (F)	56.24	68.98	83.74	82.18	95.04	101.65	2351626
IQR Max Temp (F)	2	4.984	9.499	9.904	14.217	25.334	2351626
Total Precip (Inches)	0	1.009	2.550	2.967	4.349	13.819	2351626
IQR Precip (Inches)	0	0.004724	0.034655	0.158315	0.238235	2.434279	2351626
Quantity (kGal)	0.1	2.6	4.3	6.537	7.4	1275.8	2351626
Payment (\$)	8.426	19.118	27.436	57.746	53.074	17577.749	2351626

Note: Heavy Water Appliances include pool, hot tub, sprinkler system, fountain, and car wash station/area. In the real estate industry, a full bathroom requires a sink, a tub, a shower, and a toilet. If it only has 3/4 of fixtures, it will constitute a 3/4 bathroom, and only 2/4 fixtures will constitute a half bathroom.

Quantity vs. Price Elasticity by Income Strata

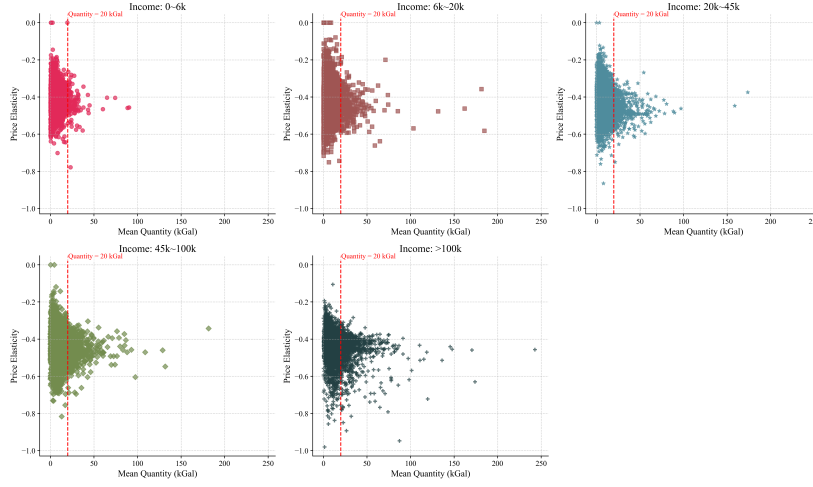


Figure 5. : Price Elasticities v. Mean Quantity by Income Strata

can be a potential mechanism explaining the findings from previous literature⁴ where they conclude that IBP optimization will likely favor higher income strata. Essentially, I would like the utilities to be aware of the price sensitivities of higher income strata, as the higher paying consumers are not actually “reliably” contributing towards the revenue, as well as the existence of the low elasticity-high quantity households in the lower income strata, as these consumers who should be benefiting from the equity goal, could potentially suffer from other competing policy goals.

One important caveat for this study (and further discussion of distributional results) is that the household income I calculated actually stemmed from the correlated house value. Even though it has been proven to be generally true⁵, it is a strong assumption to assume the same distribution of house value and household income within the same zipcode. However, later during the counterfactual analysis, I conduct a specification with only the newly built houses, such that their appraisal value will be generally higher, and still find that welfare for the lowest income stratum is much lower compared to other strata. The result can be found in Section I.D.

When specifically comparing the demand distribution between income strata, I observe the difference between the empirical cdf from the new weather pattern and the empirical cdf of the status quo weather, and the highest income stratum has larger differences, for both dry and rainy weather conditions. This showcases that the “weather elasticity” is larger in magnitude for the higher income households, which aligns with the evidence I saw from Figure ?? . Low and high variance weather conditions show similar patterns.

⁴Ruijs, Zimmermann and van den Berg (2008), Echeverri (2023), and Wichman (2024)

⁵See Zhang (2016), and Kim (2020)

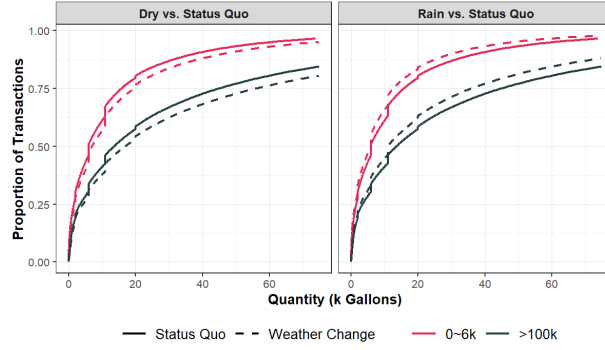


Figure 6. : Changes in Quantity Distributions between Lowest and Highest Income Strata - Dry and Rainy

The low elasticities of the low-income households will be quite concerning regarding the distributional effect, especially when the demand curve shifts rightwards. When the demand curve shifts rightwards, the conservation goal will require the price to increase at least at higher tiers. Balancing both goals with a single price will be challenging for the utilities and will certainly create a shadow cost, which will inevitably decrease welfare. Based on the counter-intuitive price elasticity results in Figure ?? and the existence of high quantity households in lower income strata in the 'Price Elasticities of Income Strata' section of the Online Appendix, it is reasonable to question if the price itself can achieve both policy goals, as well as improve the distributional effect.

C. Ramsey Pricing Model

RAMSEY PRICING MODEL WITH CONSERVATION CONSTRAINT

In this subsection, I briefly showcase the changes from the classic Ramsey model by using a simplified pricing model with the price p , chose by the water utility to maximize consumer welfare, subject to two constraints: 1) annual revenue ($R(p)$) is greater than or equal to an exogenous total annual cost (C) that is independent of demand (simplifying for large fixed costs),⁶ and 2) total annual quantity ($q(p)$) is less than or equal to an exogenous annual quantity upper bound ($\bar{Q}$).

$$(2) \quad \begin{aligned} \max_p \quad & CS(p) \\ \text{s.t.} \quad & R(p) - C \geq 0 \\ & \text{and } q(p) \leq \bar{Q} \end{aligned}$$

⁶In reality, the total cost of maintaining urban water is not related to demand fluctuation; regardless of the amount of water consumed, maintenance and transportation fees do not change, making this a reasonable assumption.

Solving this Lagrangian yields the following result:

$$(3) \quad \underbrace{\frac{p-C}{p}}_{\text{Markup}} = \underbrace{\frac{\lambda-1}{\lambda} \frac{1}{\varepsilon}}_{\text{Ramsey Rule}} + \underbrace{\frac{\mu}{\lambda p}}_{\text{Conservation Penalty}}$$

$$p^* = \frac{C + \mu/\lambda}{1 - \frac{\lambda-1}{\lambda} \frac{1}{\varepsilon}}$$

The classic Ramsey rule states that the markup is proportional to the inverse of price elasticity (ε), but the optimal markup here includes an additional term that serves as a penalty for the conservation constraint. The optimal price depends on λ (the Lagrangian multiplier for the revenue constraint), μ (the Lagrangian multiplier for the conservation constraint), the exogenous cost C , and the price elasticity ε . If μ increases, the shadow price of the conservation constraint increases, and the utility has a greater incentive to raise the markup. If $\mu = 0$, the conservation constraint is not binding, and the standard Ramsey rule is recovered. If C increases, the price will need to increase. If ε increases, demand becomes more elastic, and the price will need to decrease, which is consistent with the standard Ramsey rule. If I denote $k = \frac{\lambda-1}{\lambda}$, the price can be expressed as $p = \frac{C+\mu(1-k)}{1-\frac{k}{\varepsilon}} = \frac{C+\mu}{1-\frac{k}{\varepsilon}} - \frac{1}{k-\frac{1}{\varepsilon}}$. If λ increases, k (which is less than 1) will also increase. The first term increases. The second term, being minus a negative number since $k - \frac{1}{\varepsilon} < 0$, also increases. This means that when λ increases, the price will increase.

STATUS QUO REVENUE AS THE THRESHOLD

I will explore the validity of using the status quo revenue as the lower bound for counterfactual revenue in this section. Due to the lack of detailed cost breakdown data, it would be difficult to establish a supply-side cost model. However, by using aggregate cost information, I can make some inferences. In order to study the financial viability of Austin Water, I need to ensure that total revenue can cover total costs. However, total water service revenue is only a portion of Austin Water's overall revenue stream. Furthermore, the revenue I observe in this data represents only single-home residential water revenue, which itself is only a portion of total water service revenue. Nonetheless, water service revenue is typically the largest revenue stream ($> 70\%$), and single-home housing usage usually constitutes the largest part of total water service revenue ($> 50\%$). In particular, single-home housing usage is most susceptible to precipitation variation because multi-home housing is typically in condominiums, which do not have lawns, and commercial usage usually does not vary with changes in precipitation. Therefore, it is valid to use single-home housing to study the impact of weather, and compare the total revenue from single-home housing usage to some threshold to measure the revenue risk faced by the utility due to weather variation.

However, the lack of detailed cost information makes it hard to estimate the cost generated solely by single-home housing. The total costs from water services include operation, labor, debt service requirements, and funds transferred to the city government.

None of these can be separated and attributed solely to single-home housing usage.

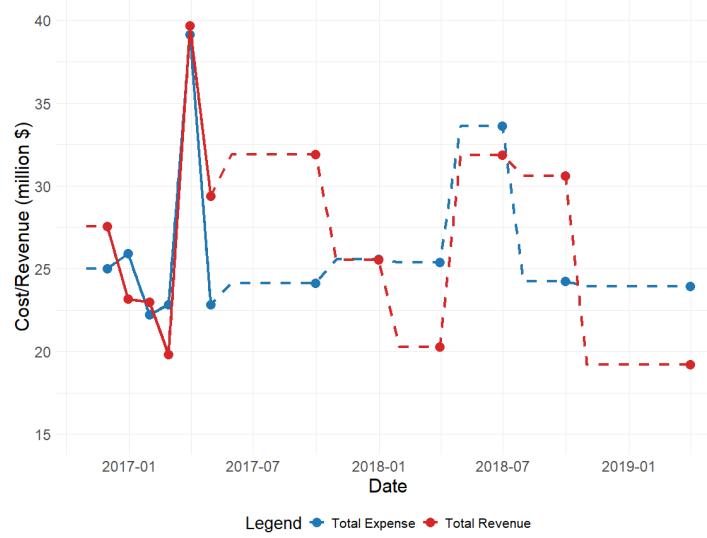


Figure 7. : Total Cost and Revenue from Water Service

Note: Austin Water releases its financial report irregularly, but on average, every 3 months. During some time between January 2017 and May 2017, they released the report every month. Hence, some data have a monthly frequency, and some do not.

Looking at the seasonal trends from quarterly financial reports of water service revenue in Figure 7 and cost, I can see that the cost for the whole year is relatively stable. The only peak is usually around the summer months when the water supply is lower and requires more water from the reservoir. The main reason for negative profit in water service is that revenues in some months fall below the typical cost level during non-peak months. Therefore, the financial risk faced by the utility is largely due to revenue volatility between months. Therefore, it is reasonable to set up a revenue lower bound for comparison with counterfactual revenues. I could set up an intricate cost model to estimate this lower bound, but the status quo revenue (which is the revenue of the year 2019, and the weather variation in 2019 was quite standard) serves as a good enough benchmark.

SIMULATE WEATHER DISTRIBUTION

It is tricky to evaluate E_Z without the knowledge of the real distribution of Z . This is due to 1) if limited to recent years of Z data, there are too few data points to empirically generate the real distribution of Z , and 2) if expanding to long-term data of Z , the weather data 30 years ago does not share the macro climate trends of the recent weather data. Unfortunately, truly predicting the weather is out of the scope of this paper, but I offer an alternative solution to estimate the optimal price based on the prior of the utility. In

addition, in one of the specifications, I used a Monte Carlo method to simulate weather prediction errors and the optimal result does not deviate that much.

The counterfactual weather is generated from a prior shift from the utility. If the utility thinks next year's weather is on average more rainy, then $\zeta_1 > 0$ and vice versa. If the utility thinks next year's weather is on average higher in variance, then $\zeta_2 > 1$. The utility will treat this counterfactual Z as the static weather and then calculate the optimal price.

Adding a Monte Carlo small log-normal disturbance will simulate the prediction error and the result is more realistic. I generate a small log-normal disturbance with $\mu = 0$ and $\sigma = 1$. The $\sigma = 1$ is calculated from the standard deviation of the strictly positive precipitation data from the recent 5 years (2014-2018). This means the utility first predicts a general trend of precipitation data, it could be either $Z' = Z \pm \zeta_1$ or $Z'/Z = \zeta_2$, and then in order to account for prediction errors, Z' is perturbed by a small log-normal error term to simulate the stochasticity of Z' based on the utility's prediction.

The Monte Carlo process is calculated as follows: for each simulation s , the utility will calculate both the objective ($CS(Z^s) - \lambda \cdot \max(0, C - R^s(Z^s))$) and the conservation constraint ($P(\sum q(Z^s) \leq \bar{Q}) \geq 0.95$) and take the average of all numbers of simulations. Essentially, the utility is maximizing over the average objective for all the simulations, while making sure the average quantity for all simulations is below the threshold. One might question why not average over all simulated months, with a total number of simulated data points to be $s \cdot 12$. It is hard to imagine the utility optimizing on a monthly basis, even though it could generate more data points. On the other hand, it is reasonable to assume that the utility would need to check the projected overall welfare and projected revenue condition by the end of their financial year. Averaging the yearly objective for all simulations essentially calculated the expected objective for the utility under the weather stochasticity. Due to the long running time, I eventually chose $s = 25$ as a start. The resulting optimal prices do not change significantly compared to Figure 9 and 10 of the main text. This showcases that the baseline model gives a consistent measurement of the first moment of weather stochasticity.

D. Counterfactual Analysis

COUNTERFACTUAL NDVI

As NDVI measures the health of vegetation, it would not make sense to keep it unchanged through shifting counterfactual precipitation. To isolate the effect of precipitation on NDVI, I conducted an OLS using the following formula, where P represents precipitation and T represents daily max temperature:

$$NDVI = \beta_0 + \beta_1 P_{\text{sum}} + \beta_2 P_{\text{iqr}} + \beta_3 T_{\text{mean}} + \beta_4 T_{\text{iqr}} + \beta_5 I$$

Note that both the mean and IQR here refer to the within-month mean and IQR. For this paper, I focus on the variance of the monthly sum of precipitation (P_{sum}); therefore, while keeping all the other variables unchanged, the parameter of interest will be β_1 . The OLS results are as follows:

Table 4—: OLS Estimation Results of Precipitation on NDVI

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.1149	0.0010	92.940	< 2e-16 ***
P_{sum}	0.0080	0.000045	177.803	< 2e-16 ***
P_{iqr}	-0.0118	0.0004	-29.460	< 2e-16 ***
T_{mean}	0.0030	0.000012	256.693	< 2e-16 ***
T_{iqr}	0.0013	0.000032	40.768	< 2e-16 ***
I (in millions)	1.449	1.3	114.940	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Multiple R-squared: 0.127, Adjusted R-squared: 0.127

F-statistic: 3.935e+04 on 5 and 1,356,918 DF, p-value: < 2e-16

This regression captures the impact of precipitation on vegetation health ecologically. The only control variable included is income to capture the size of the lawn for each household. This means for every change in precipitation in inches, the NDVI will increase by 0.008, keeping demand and income the same. Through the calculation of counterfactual NDVI, I can calculate the counterfactual demand, welfare, etc.

OPTIMAL PRICE PROCEDURE

Given the counterfactual weather and NDVI, household characteristics, and income, I will use Equation 12 of the main text to solve for the optimal price of 14 parameters. All parameters are part of IBP; therefore, I set all prices to be larger than 0, and force each step to be increasing for both marginal prices and fixed payment (meaning the difference between each tier will at least be \$0.01). The optimization algorithm I used is COBYQA, a derivative-free optimization solver designed to supersede COBYLA to solve for the bounded optimization process. In essence, COBYQA is a trust-region SQP method based on quadratic models obtained by underdetermined interpolation.⁷ Of all the constrained optimization algorithms, COBYQA consistently can provide reliable results as long as the initial value is within the valid range with respect to constraints.

Since I am dealing with price optimization, I do not require very refined results. I chose the initial searching radius of the algorithm to be 1.0, and the final radius for convergence tolerance to be 0.01 (as the lowest price difference will be in cents). I also utilize putting the loss function in the objective function in addition to the conservation constraint to avoid optimizing on 2 constraints at the same time (since otherwise, without the loss function, both constraints will need to be implemented). I choose to set the initial value as if each price jump is the same between tiers to intensify the certain benefit of more salient price jumps between certain tiers. Since COBYQA is good at finding relatively local results, I choose the initial value to be close to the status quo IBP.

⁷The algorithm is developed by Ragonneau (2022), Ragonneau and Zhang (2025)

Aside from weather stochasticity, I set up a Monte-Carlo simulation for ε using the result from Table A5 in the Online Appendix. Note that there are two levels of stochasticity with unobserved taste (η) and perception error (v). The perception error is purely random and not controlled by the household, so for each household for each month, I will generate a simulation for v_{mh} . As for η , I assume each household has an unchanged $\bar{\eta}_h$ across all months to avoid overfitting and simulate the results. I have tried using η_{mh} , the counterfactual welfare generates very little difference between specifications.

PRICE RESULT - MEAN SHIFT

Since the 5-tier IBP includes both marginal prices and fixed payments, I argue the most effective way to visualize the pricing structure is by plotting total payment (in dollars) versus quantity (in thousand gallons). Note that all plots are truncated at 50 k gallons to focus on the changes around the majority of the data points. In practice, a decent number of consumers use significantly more than 50 k gallons. In this subsection, I will only focus on ζ_1 and only showcase the result of $\zeta_1 = -0.25$ and $\zeta_1 = 0.25$. The results of non-extreme weather conditions in between fit the general trend and are omitted for clarity. All price plots include three benchmarks derived from the status quo. The status quo price (shown with a dot-dash line) reflects the current pricing structure used by Austin Water. The status quo flat price is constructed by averaging the five marginal prices and five fixed payments, resulting in a linear pricing structure. I also include the empirical cdf of the quantity to showcase the change in distribution.

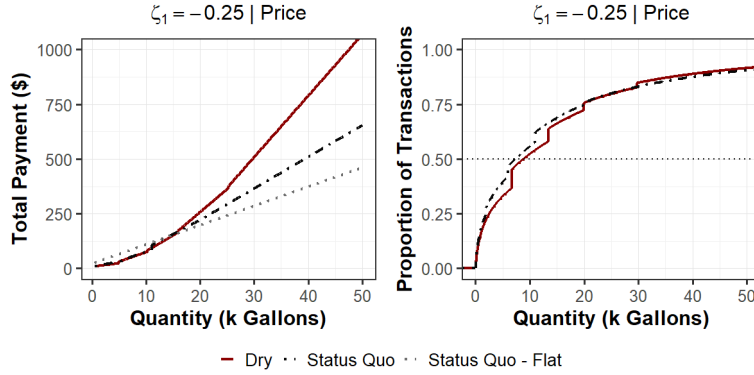
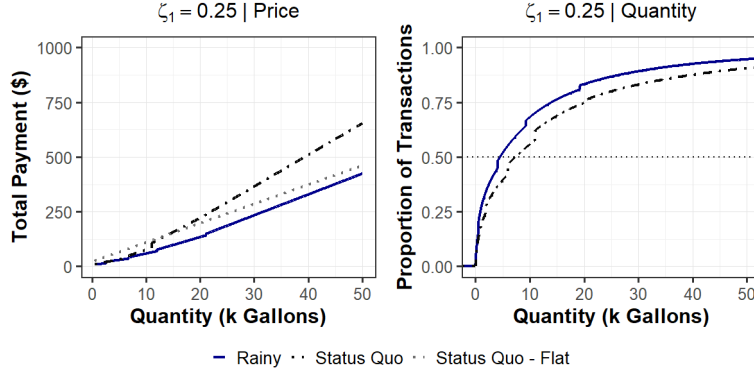


Figure 8. : Optimal Prices - $\zeta_1 = -0.25$

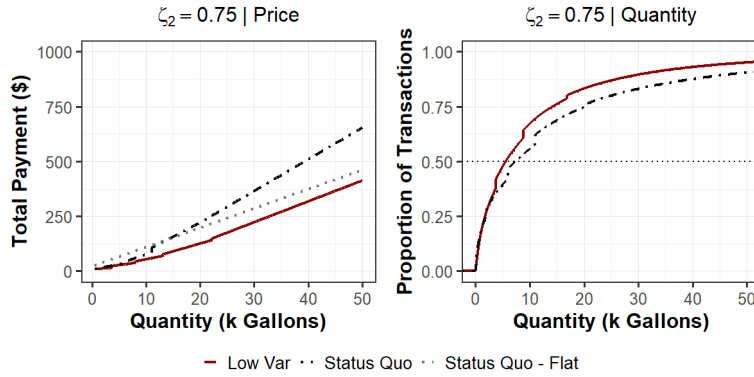
From Figure 8, when $\zeta = -0.25$, weather pushes the demand curve to the right, but the binding conservation constraint actively curb high quantity users, making the price for the higher tier much higher, and quantity distribution for higher quantity ($> 20k$ Gallons) shifts to the left. The rising price for the higher tier causes welfare loss for the high quantity users, and the existence of the high-quantity-low-income users generating the regressive welfare effect from pricing.

Figure 9. : Optimal Prices - $\zeta_1 = 0.25$

From Figure 9, when $\zeta = 0.25$, weather pushes the demand curve to the left. Since the revenue loss is less in priority for the utility ($\lambda = 0.5$), and the existence of high-quantity-low-income users push the price downward for all tiers. Even with the price decrease, the demand curve still shifts to the left, as evidenced by the empirical cdf.

PRICE RESULT - VARIANCE SHIFT

When changing ζ_2 (the ratio of weather standard deviation compared to the status quo), I will only focus on the extreme cases of $\zeta_2 = 0.75$ and $\zeta_2 = 1.25$.

Figure 10. : Optimal Prices - ζ_2

From Figure 10, when $\zeta = 0.75$, the quantity distribution shifts towards the median. However, since the status quo (and the counterfactual) quantity distribution is right-skewed, the quantity in general decreases. This will result in a similar situation of $\zeta_1 > 0$

where the revenue losses are more pressing, which will in turn result in an increase in prices across all tiers.

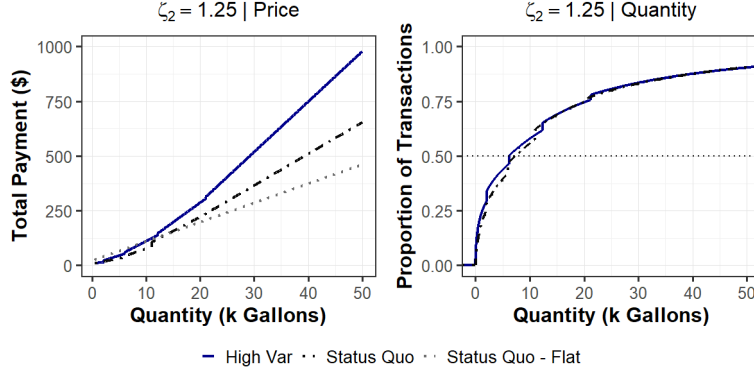


Figure 11. : Optimal Prices - ζ_2

From Figure 11, when $\zeta = 1.25$, the quantity distribution becomes more spread out. Due to the right-skewness of the data, the quantity in general increases. This will result in a similar situation of $\zeta_1 < 0$ where the conservation constraint is binding. However, what is different from the cases of $\zeta_1 < 0$ is the decreasing quantity in even drier months, making the price for lower tiers higher due to the financial risks during the now even drier months.

WELFARE RESULTS OF DIFFERENT BUILT YEARS - ROBUSTNESS CHECK

One of the concerns regarding the welfare result is that, since I am essentially equating house value to household income. There could be some mis-categorization where a household built early with a low appraisal value, but actually has a higher income. I am going to test the counterfactual analysis with the welfare from the optimal price in shifting mean and extend the result from Figure 7 of the Online Appendix with different segments of the households cohort. I will test two specifications: 1) limiting to only the houses built after 2000, which is the top 27.4% of the entire cohort, and 2) limiting to only the houses built after 1982, which is the top 50% of the entire cohort. Since the lowest income stratum will have the largest welfare swing, I will only limit the welfare result of that stratum.

When comparing to the full sample, both specifications still generate comparable welfare results, and limiting to houses with the built year in the top 50th percentile even generates lower welfare. When $\zeta_1 = -0.25$ (the demand shifts to the right), the full sample has an average EV/I for the lowest income stratum of -4.49% , while 1) limiting to only the houses built after 2000 (the 27.4 th percentile) has an average EV/I for the lowest income stratum of -3.79% , and 2) limiting to only the houses built after 1982 (the 50 th percentile) has an average EV/I for the lowest income stratum of -5.15% . The fact

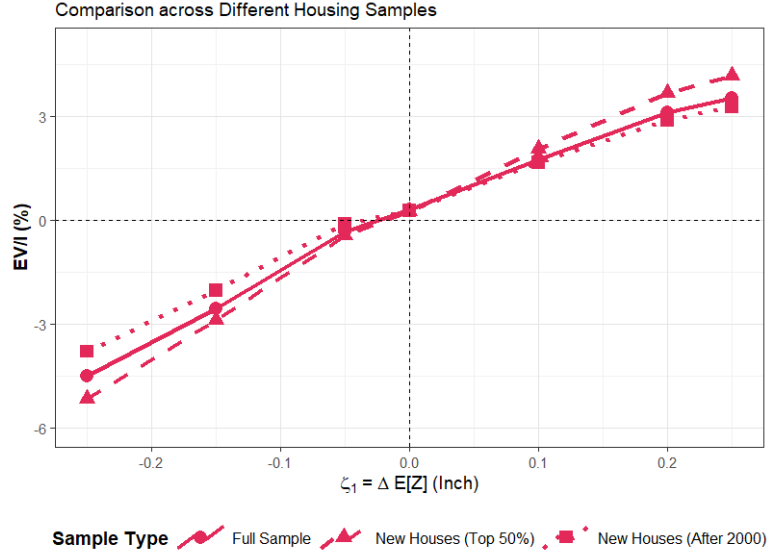


Figure 12. : Welfare from Status Quo Price in Shifting Mean ($\zeta_1 \in [-0.25, 0.25]$)

that limiting the sample to above the median generates even lower welfare, showcasing that the way of estimating income (namely, extrapolating the zip-code level income by house value distribution) is largely accurate, and the welfare result from Figure 7 of the Online Appendix is robust.

STATUS QUO PRICE - WEATHER CHANGE

In order to separate the welfare loss from weather itself and the welfare loss from the shadow cost of the policy constraints, I calculate the welfare using the status quo price vector $\mathbf{p}_0$ under all counterfactual weather conditions ($\zeta_1 \in [-0.25, 0.25]$, $\zeta_2 \in [0.75, 1.25]$). Here are the welfare result for the mean shift ($\zeta_1 \in [-0.25, 0.25]$) and variance shift ($\zeta_2 \in [0.75, 1.25]$):

I can see the clear shift of the demand curve through the weather and how it's affecting both total quantity and total payments from all strata for both cases of shifting mean and variance. Without the optimal pricing procedure considering the policy constraints, it will be difficult to curb consumption when $\zeta_1 < 0$ and lower bound the revenue when $\zeta_1 > 0$. However, the upside of not including the policy constraint is an increase in welfare, in particular, a significant increase in welfare for the lowest income stratum.

HIGH QUANTITY CONSUMERS

There are plenty of high quantity consumers, but what truly makes the equity goals of the optimal price hard to achieve is the existence of high quantity consumers in lower tiers. I have already explored the existence of these consumers in the data, but here

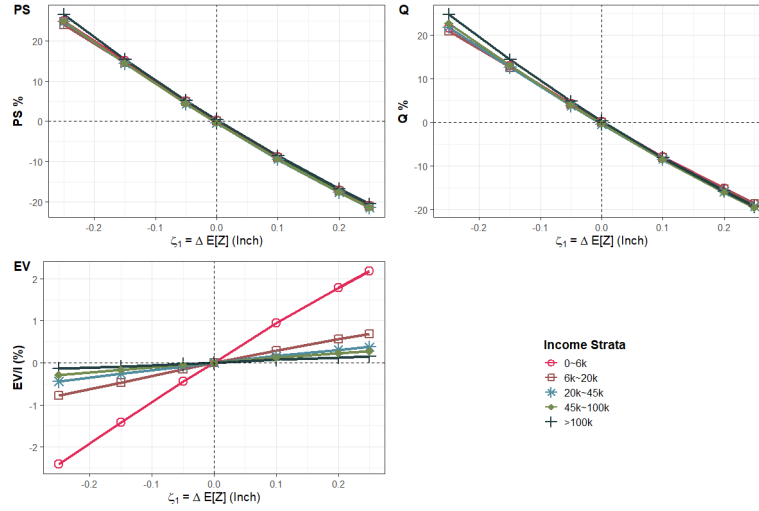


Figure 13. : Welfare from Status Quo Price in Shifting Mean ($\zeta_1 \in [-0.25, 0.25]$)

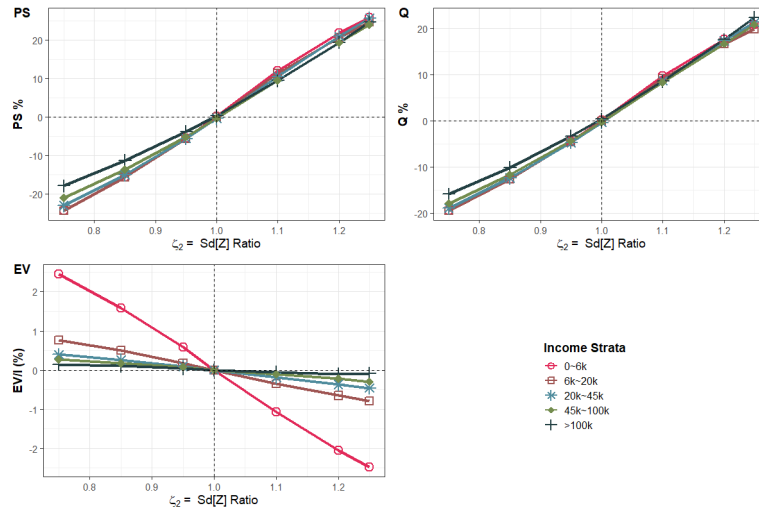


Figure 14. : Welfare from Status Quo Price in Shifting Variance ($\zeta_2 \in [0.75, 1.25]$)

I would like to specifically show the welfare loss incurred by these customers in the lowest stratum (0~6k) and compare it to the highest stratum (>100k). I will focus on the specific weather pattern of $\zeta_1 = -0.25$ where the welfare for the lowest stratum is the lowest.

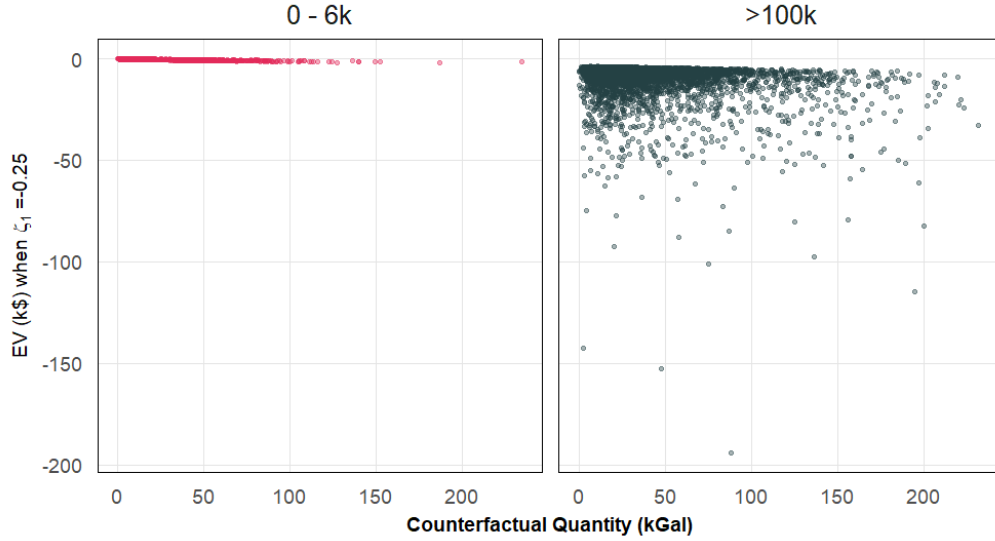


Figure 15. : EV when $\zeta_1 = -0.25$, 0 ~ 6k and >100k strata

From this picture, I can clearly see that the magnitude of welfare losses is higher in the higher income strata across different quantity levels. However, if I measure EV/I , which is a fairer measurement of welfare across income strata:

Even though the higher income stratum loses more welfare, the high quantity users (those higher than 20k gallons), and their relatively low elasticities, cause lower welfare in extreme weather conditions for the lowest income stratum.

To separate the heterogeneities of the quantity, and since income turns out to be a problematic predictor of quantity. I performed a reduced-form logit regression to figure out the source of the heterogeneity. I define high quantity transactions as > 20k gallons and only limit to the lowest income stratum. I include some predictors from the demand model, as well as variables like HVAC category (a categorical variable labeling the size of the house), lawn percentage (a continuous variable measuring the percentage of the lot size not counted as living areas), and the interactions with precipitation. Here is the

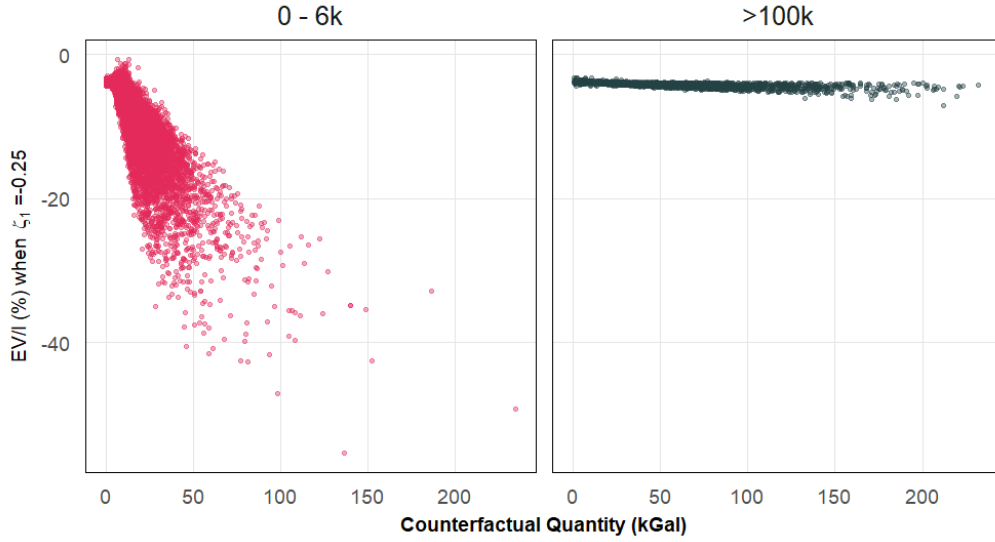


Figure 16. : EV/I when $\zeta_1 = -0.25$, $0 \sim 6k$ and $>100k$ strata

equation and the result.

$$\begin{aligned}
 \mathbb{1}\{\text{High Quantity}\} = & \text{house value} + \text{precipitation} + \text{NDVI} + \text{bathroom} + \text{bedroom} \\
 & + \text{spa area} + \text{heavy water app} + \text{HVAC category} + \text{lawn percentage} \\
 & + \text{precipitation} \times \text{house value} + \text{precipitation} \times \text{NDVI} \\
 & + \text{precipitation} \times \text{bathroom} + \text{precipitation} \times \text{bedroom} \\
 & + \text{precipitation} \times \text{spa area} + \text{precipitation} \times \text{heavy water app} \\
 & + \text{precipitation} \times \text{HVAC category} + \text{precipitation} \times \text{lawn percentage}
 \end{aligned}$$

For households in the lowest-income stratum. Higher living area, more attention to lawn (NDVI), more bathrooms, and larger pool/hot tub areas (in sqft) are all strong predictors of increased water use. House value has a negative coefficient. Based on my assumption to correlate house value with house income, and the conclusion that income is not a strong predictor of heterogeneity, this is not entirely surprising. This showcases that some households with low house value (which means low income) have very high quantities. The influence of these factors changes with precipitation: the impact of vegetation is most pronounced in dry conditions, while the effect of lawn percentage and the consumption gap between the largest and smallest homes becomes more significant in wetter climates. Therefore, for the cases of dry weather conditions, where welfare loss is the largest, NDVI is a good predictor for the heterogeneities of high water consumption,

Table 5—: Logit Regression Model of Predicting High Quantity for Low-Income Households

Main Effects				Interaction Terms			
Predictor	Est.	Std. Err.	z val.	Predictor	Est.	Std. Err.	z val.
(Intercept)	-1.031***	0.138	-7.458	Precipitation : house value	0.002***	0.000	21.639
house value (k\$)	-0.004***	0.000	-38.242	Precipitation : NDVI	-0.369**	0.125	-2.965
Precipitation (Inch)	-2.345***	0.138	-17.061	Precipitation : bathroom	-0.066***	0.017	-3.902
NDVI	5.294***	0.129	41.048	Precipitation : bedroom	0.002	0.015	0.112
bathroom	0.209***	0.019	10.892	Precipitation : spa area	-0.001***	0.000	-10.222
bedroom	-0.016	0.016	-0.996	Precipitation : heavy water app	0.110	0.152	0.723
spa area (sqft)	0.003***	0.000	13.203	Precipitation : HVAC Med-Low	-0.104**	0.033	-3.101
heavy water app (sqft)	-0.091	0.199	-0.458	Precipitation : HVAC Med-High	0.037	0.039	0.967
HVAC Med-Low	0.414***	0.032	12.944	Precipitation : HVAC High	0.128**	0.048	2.691
HVAC Med-High	0.696***	0.042	16.721	Precipitation : lawn percentage	0.472**	0.155	3.036
HVAC High	1.009***	0.057	17.739				
lawn percentage	0.212	0.156	1.361				

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1. Est. = Estimate, Std. Err. = Std. Error, z val. = z value.

and zeroscaping is a natural additional policy to remedy the impact from NDVI. Further studies could be extended towards the impact of bathrooms and larger pool/hot tub areas and their related policy impacts.

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